

Ross Kelso

Edinburgh, Scotland

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Experienced software developer with 5+ years experience in full-stack development, data engineering, and AI.

Skilled in rapid build out of efficient data analysis/visualisation systems in `c#`, `python`, and `javascript`. Driven to explore emerging technologies and frameworks.

First class degree in Computer science with a background in industrial software, machine learning, and computer vision.

Technical Skills

Languages & Frameworks: `c#`, `python`, `javascript`, `typescript`, `.net`, `angular`, `react`, `sql`, `rust`

Technologies: `elasticsearch`, `git`, `azure`, `aws`, `docker`, `kubernetes`, `tensorflow`, `nodejs`, `windows`, `linux`, `ci/cd`

Concepts: Machine Learning, Data Engineering, Cloud Architecture, Computer Vision

Work Experience

Cintra HR & Payroll (since June 2025)

Creating HR & payroll management software.

`c#` `typescript` `angular`

- Through rigorous automated testing, identified and fixed longstanding architectural issues with existing validation middleware resulting in validation accuracy and data consistency.
- Applied source generation to ensure type safe and performant interoperability between new and legacy systems.
- Delivered new pensions functionality ahead of schedule while working in a cross-functional team.
- Communicated technical and domain problems clearly and effectively.
- Designed and implemented methodology for assessing API throughput at scale. Applied gathered information to increase maximum possible throughput by 5 times.

Intelligent Plant - Software Developer (2021-2025), Intern (University Holidays 2017-2021)

Creating data storage, analysis and visualisation solutions for the energy industry.

Lead developer for the [Industrial App Store API Python Client](#) `python` `pandas` `django` `tensorflow`

- Designed and implemented a Python SDK to simplify access to the Industrial App Store API for data science environments.

- Developed comprehensive documentation and example use cases for ease of adoption by other developers.

Lead developer for [Sequence of Events Explorer](#) `c#` `javascript` `html` `css` `elasticsearch` `ci`

- Developed the application from inception to maturity, including design, development, deployment, and maintenance resulting in it being adopted by multiple major industrial clients.
- Enhanced user experience by developing an intuitive, and powerful frontend for advanced data queries.
- Improved code quality and maintainability by managing developers through rigorous code reviews and enforcing best practices.

Developer for [Alarm Analysis](#) and Intelligent Plant Historian `c#` `elasticsearch`

- Developed and maintained a backend for performing complex data aggregations.
- Worked with clients directly to facilitate on-site installations.
- Upgraded the historian across several major versions of Elastic Search. This more than doubled the size of a query that can be performed in 1s and increased the size of a query that can be performed in 5s by 10 times.

Developer for [Gestalt \(PnID & Trend\)](#) `c#` `sql` `javascript`

- Lead the project to decouple the Gestalt web app from our data API, resulting in smoother deployments for Gestalt and our API.
- Designed and implemented a database model to replace an existing file system dependency paving the way for migration to App Services and zero downtime.

Lead developer for [Power BI Industrial Connector](#) `c#` `power query m` `react`

- Developed a high-performance data connector to enable seamless industrial data access in Power BI.
- Optimized data retrieval and transformation processes to improve query efficiency and responsiveness.

Education

University of St Andrews - Computer Science MSci First Class (2016-2021)

Deans' List, for academic excellence 2017/18, 2018/19, 2019/20

Masters Thesis

[Synthetic GPS Traces for Privacy Protection](#) `python` `tensorflow` `pandas` `numpy`

- Using Markov Models and generative AI to synthesize realistic GPS traces.
- Evaluated model accuracy and effectiveness in preserving statistical properties of real-world movement data.

Senior Honours Project

[Computer Vision in Cue Sports](#) `opencv` `python`

- Designed an algorithmic computer vision system to analyze pool table positions without the need for markers.
- Implemented perspective transformation and object detection techniques for accurate ball position detection.

Junior Honours Project

Group software engineering project to create an encrypted wireless network of embedded devices `c`

`nodejs` `python` `git`

- Applied low-level networking principles to develop a secure, distributed communication protocol.
- Implemented encryption mechanisms for secure data transmission.
- Worked in an Agile environment, utilizing SCRUM methodologies for iterative development and collaboration.

Other Modules

- Algorithms and data structures `java`
- Artificial intelligence and machine learning `python` `java`
- Information Visualisation `python`
- Operating systems `c`
- Advanced programming projects `haskell` `python`

Personal Interests

Game Design `rust` `bevy` `godot`

Competed in the [GMTK Game Jam 2025](#) using `godot`. Implmented a [Tetris clone](#) in `rust` using `SDL2` including a simple software synthesizer to play the classic music. Implemented in [Flappy Bird clone](#) in `rust` using `bevy` game engine.

dowehavemilk.com `c` `nodejs`

- Created a (now-defunct) website that tracks whether shared items in my flat have ran out.
- The status could be updated from the website or from an ESP-32 on the fridge.

micro-telemetry `c` `python`

- Created a network of micro-controllers that recorded temperature data around my flat to an Intelligent Plant Historian

Commodore 64 Assembly Programming

- Wrote a variety of assembly programs for the commodore 64 to experiment with low-level computing using a simpler architecture.

Starfinder/D&D Game Master (2024-present)

- Leadership and Organization: Coordinated and led a group of players through a immersive and engaging story, ensuring a cohesive and enjoyable experience for all participants.

- Creative Problem-Solving: Designed and implemented challenging scenarios, puzzles, and encounters that required players to think critically and strategically to overcome.
- Adaptability and Improvisation: Improvised and responded to unexpected events, player choices, or rule interpretations, using creative problem-solving and diplomacy to resolve conflicts and maintain a positive game environment.

Pool Society Vice-president, Social Rep, Publicity and Fundraising Rep (2017-2021)

- Managed day-to-day operations, including equipment maintenance, weekly sessions and recruiting new members.
- Promoted the society through social media channels and organized fundraisers, including live streams of semesterly Finals Day events.
- Represented the university at several pool competitions, including 5 BUCS tournaments.